

MYC-C7Z010-20-V2

Product Brochure



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Version	author	Participants	date	Remark
V1.0	Sender		2022-05-18	First Edition
V1.0	Sender		2023-02-22	Supplementary explanation of JTAG startup configuration

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1. Overview

The MYD-C7Z010-20-V2 development board is composed of the MYC-C7Z010-20-V2 core board and the MYB-C7Z010/20

The motherboard is composed of a baseboard, and is an embedded core board with Xilinx Zynq-7010/7020 as the core. It uses Xilinx's latest Zynq-7000 All Programmable SoC platform based on 28nm process flow, tightly integrating ARM processor and FPGA architecture. The PS unit has the high performance and low power consumption characteristics of dual-core ARM Cortex-A9 MPCore, which can better meet various industrial needs in the design. The motherboard is equipped with serial port, network port, MMC/SD/SDIO card interface, ADC interface, CAN and other interfaces, supports Linux, and provides information including user manual, PDF baseboard schematic diagram, external expansion interface driver, BSP source code package, development tools, etc. It provides developers with a complete software development environment, reduces product development cycle, and realizes rapid product launch.

You can download the above information at any time at the following address: <http://down.myir-tech.com/>

During the development phase, it is recommended to use the evaluation kit MYC-C7Z010-20-V2 that comes with the core board to accelerate development. For more information on the [evaluation](http://www.myir-tech.com/product/) kit, please visit: <http://www.myir-tech.com/product/>

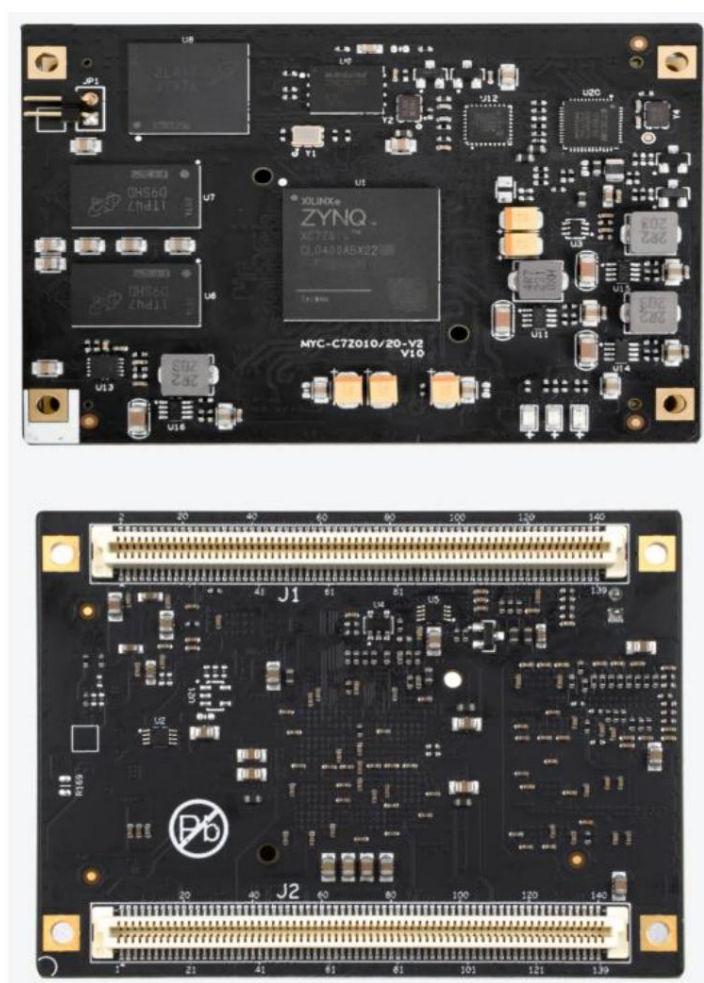


Figure 1-1 MYC-C7Z010-20-V2 core board front and back

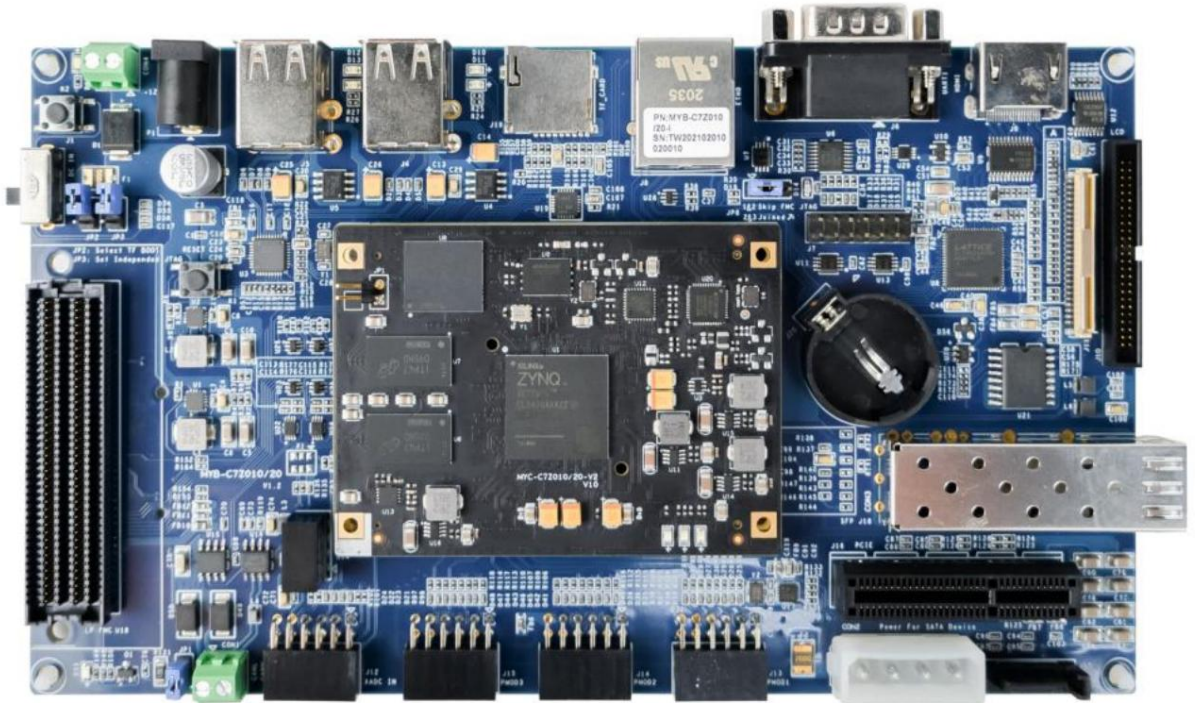


Figure 1-2 Front view of the MYC-C7Z010-20-V2 evaluation kit

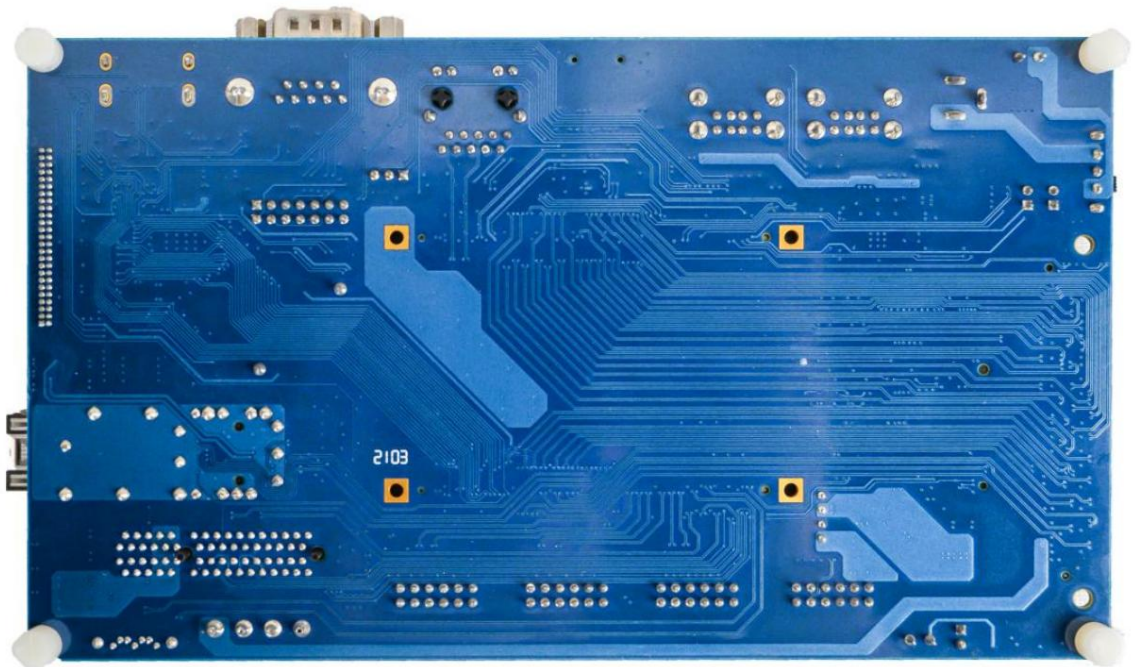


Figure 1-3 Back of the MYC-C7Z010-20-V2 evaluation kit

MYIR-MYC-C7Z010-20-V2

2. Product Introduction

The MYC-C7Z010-20-V2 core board adopts connector packaging and is equipped with a standard XC7Z010/20 processor with a main frequency of up to 667, 766MHz. It has rich peripheral interfaces and has commonly used peripheral interfaces such as USB2.0, Ethernet, UART, CAN, DCMI, ADC, etc., which is convenient for customers to flexibly customize.

The MYC-C7Z010-20-V2 series core board includes 2 specific product models: they have some differences in main frequency configuration, main CPU specifications, etc. Customers can choose the appropriate model according to their needs. For the differences between product models, please refer to the description in Section 2.4.

2.1. Chip Description

The Zynq@-7000 family is based on the Xilinx All Programmable SoC (AP SoC) architecture. These products integrate a feature-rich dual or single-core ARM Cortex-A9 MPCoreM processing system (PS) and Xilinx programmable logic (PL) on a single device, built on the most advanced, high-performance, low-power (HPL), 28 nanometer, high-k metal gate (HKMG) process technology. The ARM Cortex-A9 MPCore processor is the core of the PS, which also includes on-chip memory, external memory interfaces, and a rich set of I/O peripherals. The Zynq-7000 family offers the flexibility and scalability of an FPGA while providing the performance, power, and ease of use typically associated with ASICs and ASPs. The Zynq-7000 AP SoC family of devices enables designers to implement cost-sensitive and high-performance applications using industry-standard tools on a single platform. While each device in the Zynq-7000 family contains the same PS, the PL and I/O resources are different between devices. Therefore, Zynq-7000 AP SoC devices are able to serve a wide range of applications.

2.1.1. ZYNQ7000 Series Resource Comparison

The ZYNQ7000 series mainly includes Z-7010, Z-7020, Z-7015 etc processors.

The resource difference is reflected in whether the Coretx-A9 is single-core or dual-core, and the frequency difference. The detailed differences are shown in Table 2-1.

Feature Summary

Table 1: Zynq-7000 and Zynq-7000S SoCs

	Device Name	Z-7007S	Z-7012S	Z-7014S	Z-7010	Z-7015	Z-7020	Z-7030	Z-7035	Z-7045	Z-7100
	Part Number	XC7Z007S	XC7Z012S	XC7Z014S	XC7Z010	XC7Z015	XC7Z020	XC7Z030	XC7Z035	XC7Z045	XC7Z100
Processing System	Processor Core	Single-core ARM Cortex-A9 MPCore™ with CoreSight™			Dual-core ARM Cortex-A9 MPCore™ with CoreSight™						
	Processor Extensions	NEON™ & Single / Double Precision Floating Point for each processor									
	Maximum Frequency	667 MHz (-1); 766 MHz (-2)			667 MHz (-1); 766 MHz (-2); 866 MHz (-3)			667 MHz (-1); 800 MHz (-2); 1 GHz (-3)			667 MHz (-1) 800 MHz (-2)
	L1 Cache	32 KB Instruction, 32 KB data per processor									
	L2 Cache	512 KB									
	On-Chip Memory	256 KB									
	External Memory Support ⁽¹⁾	DDR3, DDR3L, DDR2, LPDDR2									
	External Static Memory Support ⁽¹⁾	2x Quad-SPI, NAND, NOR									
	DMA Channels	8 (4 dedicated to Programmable Logic)									
	Peripherals ⁽¹⁾	2x UART, 2x CAN 2.0B, 2x I2C, 2x SPI, 4x 32b GPIO									
	Peripherals w/ built-in DMA ⁽¹⁾	2x USB 2.0 (OTG), 2x Tri-mode Gigabit Ethernet, 2x SD/SDIO									
	Security ⁽²⁾	RSA Authentication, and AES and SHA 256-bit Decryption and Authentication for Secure Boot									
Processing System to Programmable Logic Interface Ports (Primary Interfaces & Interrupts Only)	2x AXI 32b Master 2x AXI 32-bit Slave 4x AXI 64-bit/32-bit Memory AXI 64-bit ACP 16 Interrupts										

Table 2-1 ZYNQ7000 functional diagram and resource differences of each processor

MYC-C7Z010-20-V2 processor main features:

ARM Cortex-A9, running at 667, 766MHz
16/32-bit LPDDR-2, DDR2, DDR3L, DDR3
System Level Control Registers (slcr): A set of various registers used to control the behavior of the PS
Snoop Control Unit (SCU) maintains consistency between L1 and L2
Direct Memory Access Controller: Four channels of PS (memory copies to/from any memory in the system)
256kb on-chip SRAM (OCM) with parity check: accessible to CPU, PL and central interconnect
Single/Dual Quad-SPI Controller
2 PHY, 2 USB, 2 SD, 2 SPI, 2 CAN, etc.
Static Memory Controller (SMC) Any one can be the primary boot device
Accelerator Coherence Port (ACP) from PL (master) to PS (slave)
Watchdog timer, triple counter/timer
Generic Interrupt Controller (GIC): distributes shared peripheral interrupts (SPI) from other parts of the system (PS and PL)
Package: BGA400

Table 2-2 MYC-C7Z010-20-V2 Main features

2.2. MYC-C7Z010-20-V2 Main Parameters

name	Main parameters
Main control chip series	XiLinX Zynq 7000 Series
Main control chip model	XC7Z010-1CLG400I (standard configuration) XC7Z020-1CLG400I (standard configuration)
Processor specifications	667/766MHz ARM Cortex-A9
Memory	DDR3(256Mbx16ÿ SDRAM 256Mbx16
Memory	QSPI Flash(256MbÿÿEMMC(4GB)
Core board size	75 x 55 x 8.3 mm
Interface Type	Board to Board Connector, 1x2 140Pin, 0.8mm
PCB board specifications	10-layer board design, immersion gold process
operating system	Linux 5.4.0 (latest)

Table 2-3 Main parameters

2.3. System Block Diagram

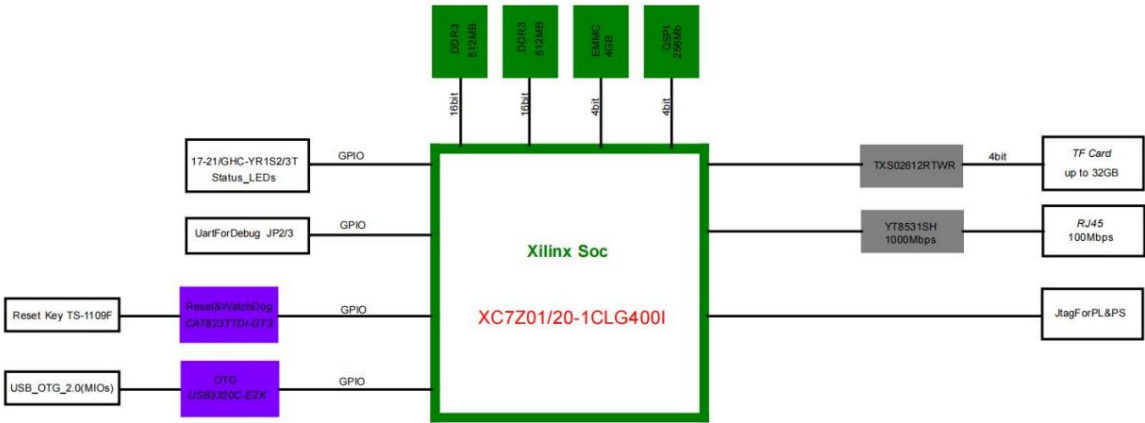


Figure 2-3 MYC-C7Z010-20-V2 core board system block diagram

MYIR-MYC-C7Z010-20-V2

2.4. Standard models

According to the different parameters such as CPU model, memory device, operating temperature, internal configuration, MYC-C7Z010-20-V

2 The core board is divided into 2 models, please choose the model that best suits you from the list below.

model Specification	MYC-C7Z010-V2-4E512D-667	MYC-C7Z020-V2-4E512D-766
Main chip	ZYNQ7000	ZYNQ7000
Main chip series	XC7Z010-1CLG400I	XC7Z020-1CLG400I
Kernel	Cortex-A9	Cortex-A9
Frequency	667MHz	766MHz
Operating system	Linux 5.4.0	Linux 5.4.0
EMMC	4G	4G
Memory	1G DDR3	1G DDR3
Memory	256Mb Flash	256Mb Flash
UART	2-way	2-way
CAN	2-way	2-way
USB / OTG 2.0		2-way
Ethernet	2-way RGMII / RMII	2-way RGMII / RMII
I2C	2-way	2-way
SPI	2-way	2-way
GPIO	54 Road (highest)	54 Road (highest)
Supply voltage:	+3.3V, 1.8V, 1.0V, 1.5V	+3.3V 1.8V 1.0V 1.5V
Mechanical dimensions:	75*55*8.3mm	75*55*8.3mm
Working temperature	-40℃ - +85℃	-40℃ - +85℃
Package pin count	400	400
Related certification	CE / ROHS	CE / ROHS

Table 2-4 MYC-C7Z010-20-V2 core board selection table 1

3.1. Pin diagram

For installation design, please refer to the instructions in Section 7.2:

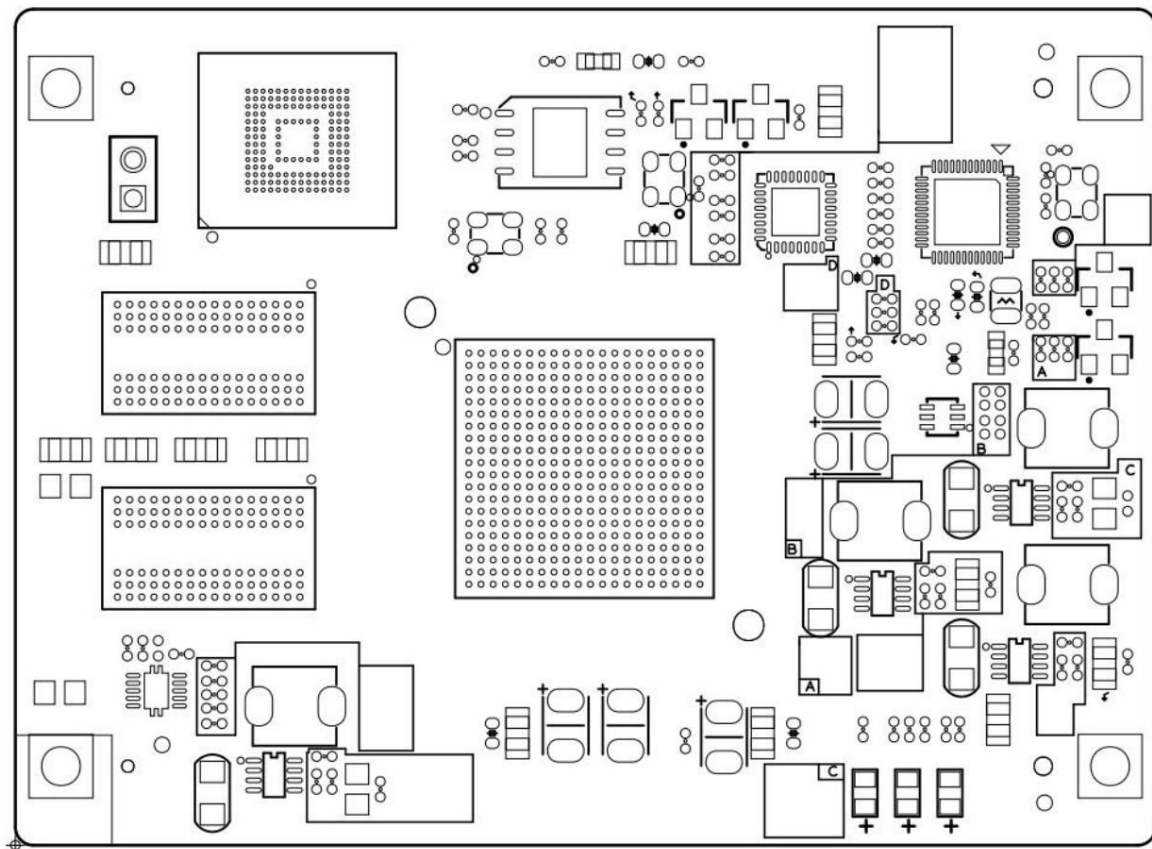


Figure 3-1 Front view of the module

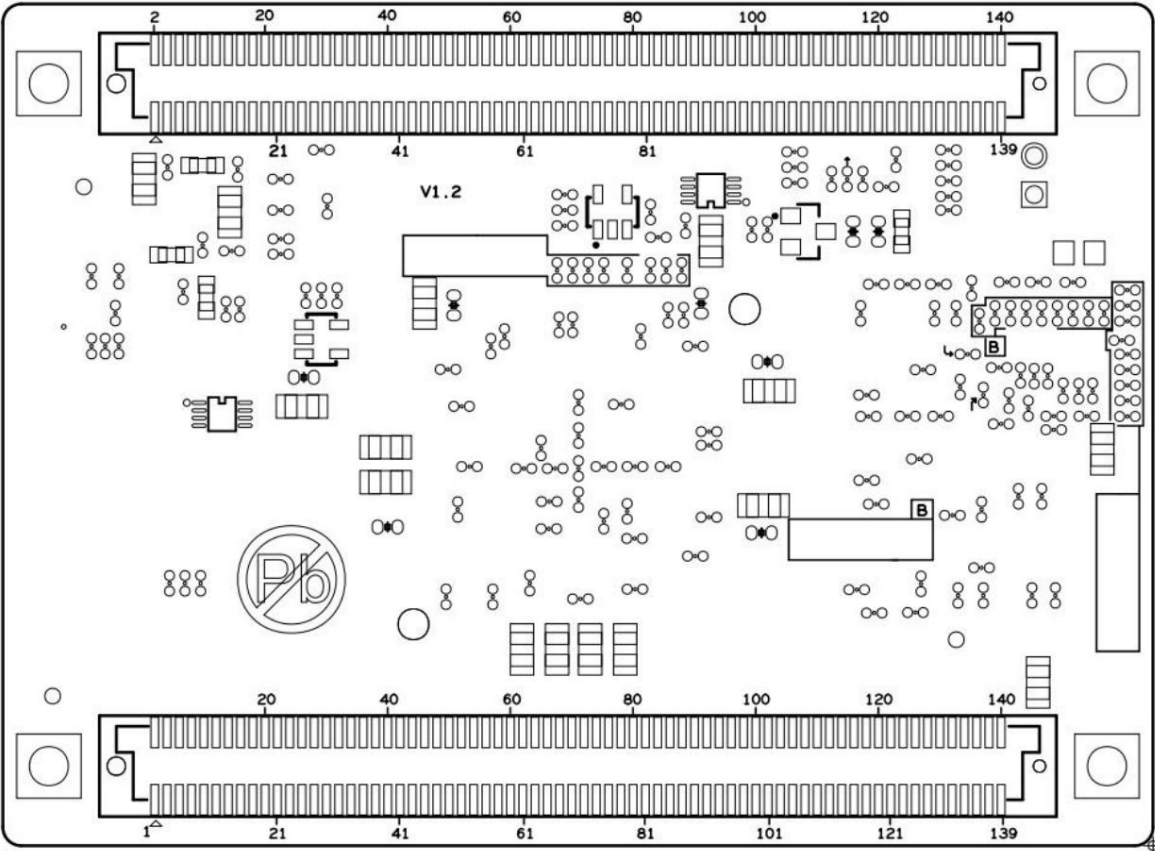


Figure 3-2 Rear view of the module

3.2. Core board pin comparison table

The interface pins of the MYC-C7Z010-20-V2 core board are defined in the following table. The pin functions of the BSP development kit are as follows:

The "default functions" in the table below are configured. If you need to change the default functions of the pins, please modify the relevant driver configuration code.

Uncertain abnormal situations such as driver conflicts may occur.

Core board pin number	Default function	Function description	Level	Input/Output	MCU Pin	Remarks
J1-P1	VDD_5V	5V power supply 5V input	5V	.	-	
J1-P2	VDD_5V	5V power supply 5V input	5V	.	-	
J1-P3	VDD_5V	5V power supply 5V input	5V	.	-	
J1-P4	VDD_5V	5V power supply 5V input	5V	.	-	
J1-P5 GND		GND Power ground	0V — —			
J1-P6 GND		GND Power ground	0V — —			
J1-P7	RGMII0_TRP0 RGMII	RGMII differential signal	1.8V	THIS —		
J1-P8 GND		GND Power ground	0V — —			
J1-P9	RGMII0_TRN0 RGMII	RGMII differential signal	1.8V	THIS —		
J1-P10 RGMII0_TRP1 RGMII		RGMII differential signal	1.8V	THIS —		
J1-P11 GND		GND Power ground	0V — —			
J1-P12 RGMII0_TRN1 RGMII		RGMII differential signal	1.8V	THIS —		
J1-P13	RGMII0_TRP2 RGMII	RGMII differential signal	1.8V	THIS —		
J1-P14 GND		GND Power ground	0V — —			
J1-P15 RGMII0_TRN2 RGMII		RGMII differential signal	1.8V	THIS —		
J1-P16 RGMII0_TRP3 RGMII		RGMII differential signal	1.8V	THIS —		
J1-P17 GND		GND Power ground	0V — —			
J1-P18 RGMII0_TRN3 RGMII		RGMII differential signal	1.8V	THIS —		
J1-P19 USB_PHY_DP USB		USB Differential Signal	1.8V	THIS —		
J1-P20 GND		GND Power ground	0V — —			
J1-P21 USB_PHY_DM USB		USB Differential Signal	1.8V	THIS —		
J1-P22 MDIO_CK	USB	USB clock signal	1.8V	.	-	
J1-P23 GND		GND Power ground	0V — —			

J1-P24	MDIO_DATA	USB	USB data signal	1.8V	The —		
J1-P25	USB_PHY_ID	USB	USB Device ID Pin	1.8V	I —		
J1-P26	RGMII0_LED_ACT RGMII		RGMII light response signal	1.8V	The —		
J1-P27	USB_PHY_VBUS	USB	USB VBUS comparator input pin	1.8V	THIS —		
J1-P28	RGMII0_LED_LINK RGMII		RGMII LED connection signal	1.8V	The —		
J1-P29	IO_B35_0	BANK35 FMC_CLK signal pin		3.3V	IO	G14	
J1-P30	VBUS_SW_EN	USB	USB 5V enable pin	1.8V	The —		
J1-P31	IO_B35_LP7	BANK35 FMC_LA17_P_CC		3.3V	IO	M19	
J1-P32	GND	GND Power	ground	0V — —			
J1-P33	IO_B35_LN7	BANK35 FMC_LA17_N_CC		3.3V	IO	M20	
J1-P34	IO_B35_LP18	BANK35 FMC_LA23_P		3.3V	IO	G19	
J1-P35	IO_B35_LP21	BANK35 FMC_LA13_P		3.3V	IO	N15	
J1-P36	IO_B35_LN18	BANK35 FMC_LA23_N		3.3V	IO	G20	
J1-P37	IO_B35_LN21	BANK35 FMC_LA13_N		3.3V	IO	N16	
J1-P38	IO_B35_LP9	BANK35 FMC_LA22_P		3.3V	IO	L19	
J1-P39	IO_B35_LP17	BANK35 FMC_LA14_P		3.3V	IO	J20	
J1-P40	IO_B35_LN9	BANK35 FMC_LA22_N		3.3V	IO	L20	
J1-P41	IO_B35_LN17	BANK35 FMC_LA14_N		3.3V	IO	H20	
J1-P42	IO_B35_LP11	BANK35 FMC_LA01_P_CC		3.3V	IO	L16	
J1-P43	GND	GND Power	ground	0V — —			
J1-P44	IO_B35_LN11	BANK35 FMC_LA01_N_CC		3.3V	IO	L17	
J1-P45	IO_B35_LP1	BANK35 FMC_LA08_P		3.3V	IO	G14	
J1-P46	GND	GND Power	ground	0V — —			
J1-P47	IO_B35_LN1	BANK35 FMC_LA08_N		3.3V	IO	C20	
J1-P48	IO_B35_LP4	BANK35 FMC_LA07_P		3.3V	IO	J20	
J1-P49	IO_B35_LP15	BANK35 FMC_LA09_P		3.3V	IO	L20	
J1-P50	IO_B35_LN4	BANK35 FMC_LA07_N		3.3V	IO	H20	
J1-P51	IO_B35_LP15	BANK35 FMC_LA07_P		3.3V	IO	L16	
J1-P52	IO_B35_LP8	BANK35 FMC_LA10_P		3.3V	IO	M17	

J1-P53 GND	GND Power	ground	0V — —			
J1-P54 IO_B35_LN8	BANK35 FMC_LA10_N		3.3V	IO	M18	
J1-P55 IO_B35_LP12	BANK35 FMC_CLK0_M2C_P		3.3V	IO	K17	
J1-P56 IO_B35_LP16	BANK35 FMC_LA24_P		3.3V	IO	G17	
J1-P57 IO_B35_LN12	BANK35 FMC_CLK0_M2C_N		3.3V	IO	K18	
J1-P58 IO_B35_N16	BANK35 FMC_LA24_N		3.3V	IO	G18	
J1-P59 VDDIO_35_PL	3.3V power supply	3.3V output	3.3V	The —		
J1-P60 VDDIO_35_PL	3.3V power supply	3.3V output	3.3V	The —		
J1-P61 GND	GND Power	ground	0V — —			
J1-P62 GND	GND Power	ground	0V — —			
J1-P63 IO_B35_LP5	BANK35 FMC_LA06_P		3.3V	IO	E18	
J1-P64 IO_B35_LP10	BANK35 FMC_LA02_P		3.3V	IO	K19	
J1-P65 IO_B35_LN5	BANK35 FMC_LA06_N		3.3V	IO	E19	
J1-P66 IO_B35_LN10	BANK35 FMC_LA02_N		3.3V	IO	J19	
J1-P67 IO_B35_LP3	BANK35 FMC_LA03_P		3.3V	IO	E17	
J1-P68 IO_B35_LP14	BANK35 FMC_LA00_P_CC		3.3V	IO	J18	
J1-P69 IO_B35_LN3	BANK35 FMC_LA03_N		3.3V	IO	D18	
J1-P70 IO_B35_LN14	BANK35 FMC_LA00_N_CC		3.3V	IO	H18	
J1-P71 IO_B35_LP2	BANK35 FMC_LA04_P		3.3V	IO	B19	
J1-P72 IO_B35_LP6	BANK35 FMC_LA16_P		3.3V	IO	F16	
J1-P73 IO_B35_LN2	BANK35 FMC_LA04_N		3.3V	IO	A20	
J1-P74 IO_B35_LN6	BANK35 FMC_LA16_N		3.3V	IO	F17	
J1-P75 GND	GND Power	ground	0V — —			
J1-P76 GND	GND Power	ground	0V — —			
J1-P77 IO_B35_LP13	BANK35 FMC_CLK0_C2M_P		3.3V	IO	H16	
J1-P78 IO_B35_LP24	BANK35 FMC_LA15_P		3.3V	IO	K16	
J1-P79 IO_B35_LN13	BANK35 FMC_CLK0_C2M_N		3.3V	IO	H17	
J1-P80 IO_B35_LN24	BANK35 FMC_LA15_N		3.3V	IO	J16	

J1-P81 GND	GND Power ground	0V — —			
J1-P82 IO_B35_LP20	BANK35 FMC_LA20_P	3.3V IO		K14	
J1-P83 IO_B35_LP19	BANK35 FMC_LA12_P	3.3V IO	H15		
J1-P84 IO_B35_LN20	BANK35 FMC_LA20_N	3.3V IO		J14	
J1-P85 IO_B35_LN19	BANK35 FMC_LA12_N	3.3V IO	G15		
J1-P86 IO_B35_LP22	BANK35 FMC_LA19_P	3.3V IO		L14	
J1-P87 IO_B35_LP23	BANK35 FMC_LA11_P	3.3V IO	M14		
J1-P88 IO_B35_LN22	BANK35 FMC_LA19_N	3.3V IO		L15	
J1-P89 IO_B35_LN23	BANK35 FMC_LA11_N	3.3V IO	M15		
J1-P90 IO_B35_25	BANK35 FMC_SDA signal	3.3V IO		J15	
J1-P91 GND	GND Power ground	0V — —			
J1-P92 GND	GND Power ground	0V — —			
J1-P93 XADC_VCC	1.8V power supply 1.8V output	1.8V O —			
J1-P94 JTAG_TCK	BANK0 JTAG_Clock signal	3.3V IO		F9	
J1-P95 XADC_INP0	BANK0 XADC differential signal line	1.8 V I		K9	
J1-P96 JTAG_TMS	BANK0 JTAG_Chip Select Signal	3.3V IO		J6	
J1-P97 XADC_INN0	BANK0 XADC differential signal line	1.8 V I		L10	
J1-P98 JTAG_TDI	BANK0 JTAG_Input Signal	3.3V IO	G6		
J1-P99 XADC_TEMP_P	BANK0 XADC differential signal line	1.8 V I		M9	
J1-P100 JTAG_TDO	BANK0 JTAG_OUTPUT signal	3.3V IO		F6	
J1-P101 XADC_TEMP_N	BANK0 XADC differential signal line	1.8 V I		M10	
J1-P102 GND	GND Power ground	0V — —			
J1-P103 XADC_GND	XADC_GND Power Ground	0V — —			
J1-P104 SDIO0_CLK	SD Card SD Card Clock Signal	1.8V O	D14		
J1-P105 GND	GND Power ground	0V — —			
J1-P106 SDIO0_CMD	SD card SD card command signal	1.8V O	C17		
J1-P107 PS_MIO0	BANK500 WDT_FEED	3.3V O	E6		
J1-P108 SDIO0_D1	SD card SD card data signal IO port	1.8V I		E12	

J1-P109	\$_JTAG_H	BANK500 mode start pin	3.3 VI		A6	
J1-P110	\$DIO0_D2	SD card SD card data signal IO port	1.8V IO	A9		
J1-P111	\$D_BOOT_CONF_H	BANK500 mode boot pin	3.3 VI		B7	
J1-P112	\$DIO0_D3	SD card SD card data signal IO port	1.8V I		F13	
J1-P113	PS_MIO8	CAN CAN TX signal	3.3V O	D5		
J1-P114	\$DIO0_D4	SD card SD card data signal IO port	1.8V I		B15	
J1-P115	PS_MIO9	CAN CAN RX signal	3.3VO	B5		
J1-P116	\$DIO0_CD	SD card SD card data signal IO port	1.8V IO	D16		
J1-P117	PS_MIO10	EMMC EMMC data signal	3.3V IO		E9	
J1-P118	\$DIO0_WP	SD card SD card data signal IO port	1.8V I		B14	
J1-P119	PS_MIO11	EMMC EMMC chip select signal	3.3V IO		C6	
J1-P120	GND	GND Power ground	0V —			
J1-P121	PS_MIO12	EMMC EMMC clock signal	3.3V IO	D9		
J1-P122	UART1_TX	UART UART sends signal	1.8V O	B12		
J1-P123	PS_MIO13	EMMC EMMC data signal	3.3V IO		E8	
J1-P124	UART1_RX	UART UART receive signal	1.8 VI		C12	
J1-P125	PS_MIO14	EMMC EMMC data signal	3.3V IO		C5	
J1-P126	PS_501_MIO_50	BANK500 I2C clock signal	1.8 VI		B13	
J1-P127	PS_MIO15	EMMC EMMC data signal	3.3V IO		C8	
J1-P128	PS_501_MY_51	BANK500 I2C data signal	1.8VO	B9		
J1-P129	GND	GND Power ground	0V —			
J1-P130	FPGA_INITn	BANK0 FPGA_INITn signal	3.3VO	R10		
J1-P131	Sys_PG	Power PG System input power PG signal 5V		0 —		
J1-P132	FPGA_WARMRESETn	BANK0 FPGA wake-up signal	3.3 VI		L6	
J1-P133	PS_500_RESET_OUTn	BANK500 reset output	1.8VO	B7		
J1-P134	FPGA_INIT_DONE	BANK0 FPGA DONE signal	3.3VO	R11		
J1-P135	PS_WARMRESET_INn	BANK500 reset button signal input	1.8VI			
J1-P136	VBAT_KEY_BACK_1.8V	BANK0 VBAT_1.8V power supply	1.8V O	F11		

J1-P137	eMMC_RSTn	EMMC EMMC reset signal	3.3V	I —		
J1-P138	VDDIO_501_PS	IO level of BANK501 PS	1.8V	O E11		
J1-P139	VDDIO_500_PS	IO level of BANK500 PS	3.3V	O C7		
J1-P140	GND	GND Power ground	0V	— —		
J2-P1	VDD_5V	5V power supply 5V input	5V	I —		
J2-P2	VDD_5V	5V power supply 5V input	5V	I —		
J2-P3	VDD_5V	5V power supply 5V input	5V	I —		
J2-P4	VDD_5V	5V power supply 5V input	5V	I —		
J2-P5	GND	GND Power ground	0V	— —		
J2-P6	GND	GND Power ground	0V	— —		
J2-P7	IO_B34_25	BANK 34 power supply enable high level valid	3.3V	O T19		
J2-P8	IO_B34_0	BANK 34 backlight power PWM control	3.3V	O R19		
J2-P9	IO_B34_LP13	BANK 34 Display data enable	3.3V	O N18		
J2-P10	IO_B34_LP24	BANK 34 Capacitive touch reset	3.3V	O P15		
J2-P11	IO_B34_LN13	BANK 34 display pixel clock signal	3.3V	O P19		
J2-P12	IO_B34_LN24	BANK 34 Capacitive touch interrupt	3.3V	O P16		
J2-P13	IO_B34_LP22	BANK 34 Display data B14	3.3V	O W18		
J2-P14	IO_B34_LP21	BANK 34 display field signal	3.3V	O V17		
J2-P15	IO_B34_LN22	BANK 34 Display data B15	3.3V	O W19		
J2-P16	IO_B34_LN21	BANK 34 Display line signal	3.3V	O V18		
J2-P17	IO_B34_LP23	BANK 34 Display data B10	3.3V	O N17		
J2-P18	IO_B34_LP20	BANK 34 Display data B11	3.3V	O T17		
J2-P19	IO_B34_LN23	BANK 34 Display data B12	3.3V	O P18		
J2-P20	IO_B34_LN20	BANK 34 Display data B13	3.3V	O R18		
J2-P21	GND	GND Power ground	0V	— —		
J2-P22	GND	GND Power ground	0V	— —		
J2-P23	IO_B34_LP19	BANK 34 Display data B8	3.3V	O R16		
J2-P24	IO_B34_LP18	BANK 34 Display data B6	3.3V	O V16		
J2-P25	IO_B34_LN19	BANK 34 Display data B9	3.3V	O W16		

J2-P26 IO_B34_LN18	BANK 34 Display data B7	3.3VO R17		
J2-P27 IO_B34_LP9	BANK 34 Display data B4	3.3VO T16		
J2-P28 IO_B34_LP12	BANK 34 Display data B2	3.3VO U18		
J2-P29 IO_B34_LN9	BANK 34 Display data B5	3.3VO U17		
J2-P30 IO_B34_LN12	BANK 34 Display data B3	3.3VO U19		
J2-P31 IO_B34_LP5	BANK 34 Display data B0	3.3VO T14		
J2-P32 IO_B34_LP14	BANK 34 FMC_LA21_P	3.3V IO N20		
J2-P33 IO_B34_LN5	BANK 34 Display data B1	3.3V IO T15		
J2-P34 IO_B34_LN14	BANK 34 FMC_LA21_N	3.3V O P20		
J2-P35 VDDIO_34_PL	BANK 34 VDDIO Level	3.3VO —		
J2-P36 VDDIO_34_PL	BANK 34 VDDIO Level	3.3VO —		
J2-P37 GND	GND Power ground	0V — —		
J2-P38 GND	GND Power ground	0V — —		
J2-P39 IO_B34_LP11	BANK 34 FMC_LA18_P_CC	3.3V IO U14		
J2-P40 IO_B34_LP15	BANK 34 FMC_LA25_P	3.3V IO T20		
J2-P41 IO_B34_LN11	BANK 34 FMC_LA18_N_CC	3.3V IO U15		
J2-P42 IO_B34_LN15	BANK 34 FMC_LA25_N	3.3V IO U20		
J2-P43 IO_B34_LP6	BANK 34 FMC_LA32_P	3.3V IO	P14	
J2-P44 IO_B34_LP10	BANK 34 FMC_LA26_P	3.3V IO V15		
J2-P45 IO_B34_LN6	BANK 34 FMC_LA32_N	3.3V IO	R14	
J2-P46 IO_B34_LN10	BANK 34 FMC_LA26_N	3.3V IO W15		
J2-P47 IO_B34_LP4	BANK 34 FMC_LA33_P	3.3V IO V12		
J2-P48 IO_B34_LP16	BANK 34 FMC_LA24_P	3.3V IO V20		
J2-P49 IO_B34_LN4	BANK 34 FMC_LA33_N	3.3V IO W13		
J2-P50 IO_B34_LN16	BANK 34 FMC_LA24_N	3.3V IO W20		
J2-P51 GND	GND Power ground	0V — —		
J2-P52 GND	GND Power ground	0V — —		
J2-P53 IO_B34_LP7	BANK 34 FMC_LA30_P	3.3V IO Y16		
J2-P54 IO_B34_LP17	BANK 34 FMC_LA29_P	3.3V IO Y18		

J2-P55 IO_B34_LN7	BANK 34 FMC_LA30_N	3.3V IO	Y17		
J2-P56 IO_B34_LN17	BANK 34 FMC_LA29_N	3.3V IO	Y19		
J2-P57 IO_B34_LP3	BANK 34 FMC_PRSNT_M2C_L	3.3V IO	U13		
J2-P58 IO_B34_LP8	BANK 34 FMC_LA27_P	3.3V IO	W14		
J2-P59 IO_B34_LN3	BANK 34 HDMI_INT	3.3V IO	V13		
J2-P60 IO_B34_LN8	BANK 34 FMC_LA27_N	3.3V IO	Y14		
J2-P61 IO_B34_LP1	BANK 34 FMC_LA31_P	3.3V IO		T11	
J2-P62 IO_B34_LP2	BANK 34 FMC_LA28_P	3.3V IO	T12		
J2-P63 IO_B34_LN1	BANK 34 FMC_LA31_N	3.3V IO	T10		
J2-P64 IO_B34_LN2	BANK 34 FMC_LA28_N	3.3V IO	U12		
J2-P65 GND	GND Power ground	0V	—		
J2-P66 GND	GND Power ground	0V	—		
J2-P67 IO_B13_LP3	BANK 13 PCIE_SMBCLK clock	3.3V I	—		
J2-P68 IO_B13_LP7	BANK 13 SFP_DISABLED0 Shutdown transmit	3.3V IO	—		
J2-P69 IO_B13_LN3	BANK 13 PCIE_SMBDAT data	3.3V IO	—		
J2-P70 IO_B13_LN7	BANK 13 SFP_LOS0 LOS alarm	3.3V IO	—		
J2-P71 IO_B13_LP4	BANK 13 PCIE hot plug presence detection	3.3V O	—		
J2-P72 IO_B13_LP8	BANK 13 PCIE rate selection	3.3V IO	—		
J2-P73 IO_B13_LN4	BANK 13 Link Activation Wake-up	3.3V IO	—		
J2-P74 IO_B13_LN8	BANK 13 module definition pin, I2C communication data line	3.3V IO	—		
J2-P75 IO_B13_LP5	BANK 13 hot plug detection	3.3V I	—		
J2-P76 IO_B13_LP9	BANK 13 module definition pin, I2C communication data line	3.3V IO	—		
J2-P77 IO_B13_LN5	BANK 13 PCIE transmission part error	3.3V IO	—		
J2-P78 IO_B13_LN9	BANK 13 module definition pin, ground	3.3V IO	—		
J2-P79 GND	GND Power ground	0V	—		
J2-P80 GND	GND Power ground	0V	—		
J2-P81 VDDIO_13_PL	BANK 34 VDDIO Level	3.3V O	—		
J2-P82 VDDIO_13_PL	BANK 34 VDDIO Level	3.3V O	—		
J2-P83 IO_B13_LP21	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO		V11	

J2-P84 IO_B13_0	BANK 13 Expansion IO	3.3V IO V5		
J2-P85 IO_B13_LN21	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO V10		
J2-P86 IO_B13_LP16	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO W10		
J2-P87 IO_B13_LP12	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO T9		
J2-P88 IO_B13_LN16	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO W9		
J2-P89 IO_B13_LN12	BANK 13 XC7Z020 Pmod expansion IO	3.3V IO U10 only		
J2-P90 IO_B13_LP20	BANK 13 XC7Z020 Pmod expansion IO	3.3V IO Y12 only		
J2-P91 IO_B13_LP17	BANK 13 XC7Z020 Pmod expansion IO	3.3V IO U9 only		
J2-P92 IO_B13_LN20	BANK 13 XC7Z020 Pmod expansion IO	3.3V IO Y13 only		
J2-P93 IO_B13_LN17	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO U8		
J2-P94 IO_B13_LP18	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO W11		
J2-P95 GND	GND Power ground	0V — —		
J2-P96 IO_B13_LN18	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO	Y11	
J2-P97 IO_B13_LP11	BANK 13 XC7Z020 Pmod expansion IO	3.3V IO U7 only		
J2-P98 GND	GND Power ground	0V — —		
J2-P99 IO_B13_LN11	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO V7		
J2-P100 IO_B13_LP14	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO Y9		
J2-P101 IO_B13_LP15	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO V8		
J2-P102 IO_B13_LN14	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO Y8		
J2-P103 IO_B13_LN15	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO W8		
J2-P104 IO_B13_LP13	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO Y7		
J2-P105 IO_B13_LP22	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO V6		
J2-P106 IO_B13_LN13	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO Y6		
J2-P107 IO_B13_LN22	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO W6		
J2-P108 IO_B13_LP19	BANK 13 only XC7Z020 Pmod expansion IO	3.3V IO T5		
J2-P109 GND	GND Power ground	0V — —		
J2-P110 IO_B13_LN19	BANK 13 XC7Z020 Pmod expansion IO	3.3V IO U5		
J2-P111 MGTPTXP0	PCIE sends differential transmission signals	3.3V IO —		
J2-P112 GND	GND Power ground	0V — —		

J2-P113	MGTPTXN0	PCIE sends differential transmission signals	3.3V IO	—		
J2-P114	MGTPTXP2	PCIE sends differential transmission signals	3.3V IO	—		
J2-P115	GND	GND Power ground	0V	—		
J2-P116	MGTPTXN2	PCIE sends differential transmission signals	3.3V IO	—		
J2-P117	MGTPRXP0	PCIE receives differential transmission signals	3.3V IO	—		
J2-P118	GND	GND Power ground	0V	—		
J2-P119	MGTPRXN0	PCIE receives differential transmission signals	3.3V IO	—		
J2-P120	MGTPRXP2	PCIE receives differential transmission signals	3.3V IO	—		
J2-P121	GND	GND Power ground	0V	—		
J2-P122	MGTPRXN2	PCIE receives differential transmission signals	3.3V IO	—		
J2-P123	MGTPTXP1	PCIE sends differential transmission signals	3.3V IO	—		
J2-P124	GND	GND Power ground	0V	—		
J2-P125	MGTPTXN1	PCIE sends differential transmission signals	3.3V IO	—		
J2-P126	MGTPTXP3	PCIE sends differential transmission signals	3.3V IO	—		
J2-P127	GND	GND Power ground	0V	—		
J2-P128	MGTPTXN3	PCIE sends differential transmission signals	3.3V IO	—		
J2-P129	MGTPRXP1	PCIE receives differential transmission signals	3.3V IO	—		
J2-P130	GND	GND Power ground	0V	—		
J2-P131	MGTPRXN1	PCIE receives differential transmission signals	3.3V IO	—		
J2-P132	MGTPRXP3	PCIE receives differential transmission signals	3.3V IO	—		
J2-P133	GND	GND Power ground	0V	—		
J2-P134	MGTPRXN3	PCIE receives differential transmission signals	3.3V IO	—		
J2-P135	MGTREFCLKP0	PCIE PCIE reference clock positive differential signal	3.3V IO	—		
J2-P136	GND	GND Power ground	0V	—		
J2-P137	MGTREFCLKN0	PCIE PCIE reference clock negative differential signal	3.3V IO	—		
J2-P138	MGTREFCLKP1	PCIE 100M clock positive differential signal	3.3V IO	—		
J2-P139	GND	GND Power ground	0V	—		
J2-P140	MGTREFCLKN1	PCIE 100M clock negative differential signal	3.3V IO	—		

Table 3-1 MYC-C7Z010-20-V2 core board PIN LIST

4. Electrical characteristics

4.1. Main power supply (VDD_5V)

The main power supply of the MYC-C7Z010-20-V2 core board is VDD_5V, corresponding to 1,

To ensure normal operation, the baseboard must provide a voltage of $5V \pm 5\%$ and ensure that the input voltage of the power supply circuit is

The output capacity can meet the power consumption of the core board. Section 4.3 lists the power consumption and current of the core board under various conditions.

Please reserve appropriate margin when connecting the circuit. If you have high requirements for heat dissipation, you can use a radiator.

4.2. Power Domain

The external power supply voltage requires the baseboard to provide the corresponding voltage. The internal voltage is the voltage generated by the core board itself.

Additional power supply is required.

The core board is powered by VDD_5V, and the power management chip generates multiple different voltages to meet the needs of MPU, DDR3, FI
ash and other modules. The complete power chip model is MP2143DJ-LF-Z.

name	describe	Recommended voltage
VDD_5V	5V input, main supply voltage	5V

Table 4-1 External power supply voltage

4.3. Power consumption

Working conditions	Supply voltage (V)	Average current (mA)	Peak current (mA)	Total power consumption (mW)
During boot	5	200	250	1000
Full-load Stage 5		280	290	1400

Table 4-2 Power consumption parameters

4.4. GPIO DC Characteristics

parameter	Label	Minimum Value	Recommended Value	Maximum Value	Unit	Description
High level input voltage	V _{IH}	2.3	3.3	3.6	V	—
1.8V input voltage	H _{IV}	1.79V	1.8V	1.81V	V	—
Low level input voltage	V _{IL}	-0.3	—	1.0	V	—
High level output voltage	V _{OH}	—	3.3	—	V	In —
Low level output voltage	V _{OL}	—	—	0.9	V	—

Table 4-3 GPIO DC characteristics

5. System Configuration and Startup

5.1. BOOT mode settings

When the MYC-C7Z010-20-V2 series processor starts up, it will first execute the program in the Boot ROM inside the chip.

When the Boot ROM starts, it reads the BOOT pin to enter different startup sources. The specific correspondence is as follows:

Boot Pin 1	Boot Pin 2	Initial Boot Source	illustrate
0	0	JTAG	Set JTAG mode to start
0	1	QSPI	Default
1	1	Micro SD	Set Micro SD card to boot

Table 5-1 Processor boot mode configuration

There are two pins that determine the boot source. Boot Pin (1:2) are SD_BOOT_CONF_H (PS_MIO

4), S_JTAG_H (PS_MIO5) pins. These pins have pull-up or pull-down design added in the core board.

Note: When using the supporting development board, you need to manually turn on the resistor on the back of the core board to start the JTAG mode of the V12 version.

R12 is replaced on R10, jumper caps JP2 and JP3 are not closed. V10 version does not need to replace resistors, jumper caps are not closed

That's it.

5.2. Reset and switch

The MYC-C7Z010-20-V2 core board provides two dedicated pins, PS_WARMRESET_INn and

WDT_RESETN, the two have different functions, it is recommended to connect them both for different purposes.

Pin Definition	illustrate
WDT_RESETN	Power-off reset pin. You can use RC reset circuit or hardware watchdog reset chip reset input out.
PS_WARMRESET_INn and the base can be	reset with a button, and an ESD diode can be connected in parallel to eliminate static electricity.

Table 5-2 Reset and PWRON pin function description

6. Interface Description

The MYC-C7Z010-20-V2 core board will consider the size of the PCB at the beginning of the design.

Lead out all IO resources. A small amount of GPIO is discarded. In addition, specific functions such as clock, eMMC, QSPI-

Flash will also occupy some IO, etc. These IO are not led out to the core board interface. Please refer to Section 6.13 to understand the internal resource allocation of the core board.

It is recommended that users use the Xilinx data sheet and related software to allocate pin resources.

You can visit the Xilinx official website.

6.1. SDMMC Interface

The MYC-C7Z010-20-V2 core board supports the MMC interface. The chip TXS02612 can be used for expansion, a single SDIO port to interface with two SDIO peripherals. It is recommended to use the MMC1 interface for Micro SD circuit, MMC2 is used with SDIO WIFI or other SDIO interface peripherals.

6.1.1. Pin Definition

MCU pin label	default function	Functional Description	Level	Input/Output out	Core board lead foot	Remark
D14	SDIO0_CLK	SD card SD card clock signal	1.8 VO		J1-P104	
C17	SDIO0_CMD	SD card SD card command/reply signal I/O port 1.8VO			J1-P106	
E12	SDIO0_D1	SD card SD card data signal I/O port	1.8V	IO	J1-P108	
A9	SDIO0_D2	SD card SD card data signal I/O port	1.8V	IO	J1-P110	
F13	SDIO0_D3	SD card SD card data signal I/O port	1.8V	IO	J1-P112	
B15	SDIO0_D4	SD card SD card data signal I/O port	1.8V	IO	J1-P114	
D16	SDIO0_CD	SD card SD card monitoring data bit I/O port 3.3V		IO	J1-P116	
B14	SDIO0_WP	SD card SD card write protection signal I/O port 3.3V		IO	J1-P118	

Table 6-1 MMC3 interface PIN definition

6.2. UART Interface

The MYC-C7Z010-20-V2 core board has an asynchronous serial port with up to nine pins.

The core board is configured with a 3-pin serial port by default, of which UART1 has flow control (RTS and CTS signals) function.
able.

6.2.1. Pin Definition

MCU Pin No.	Default Function	Function Description	Level	Input/Output	Core Board Pin	Remarks
B12	UART1_TX	UART	UART transmission signal line 1.8VO		J1-P122	
C12	UART1_RX	UART	UART receive signal line 1.8VI		J1-P124	

Table 6-2 UART interface PIN definition

6.3. USB interface

MYC-C7Z010-20-V2 core board provides two high-speed USB2.0 controllers, which can be configured as two-way HOST Interface or one high-speed USB2.0 HOST, one high-speed USB 2.0 OTG. Please use USB1 port to design USB 2.0 OTG. Using USB transceiver: USB3320C, the chip on the core board is connected to the USB on the bottom board through it seat.

6.3.1. Pin Definition

MCU Pin Number	Default Function	Function Description	Level input/output	core board pinout	
C18	PS_MIO39 USB1	USB1-HS interface D7	1.8V	IO	
E16	PS_MIO31	USB1	USB1-HS interface NXT	1.8V	
C16	PS_MIO28 USB1	USB1-HS interface D4	1.8V	IO	
C15	PS_MIO30 USB1	USB1-HS interface STP	1.8V	IO	
A14	PS_MIO32 USB1	USB1-HS interface D0	1.8V	IO	
D15	PS_MIO33 USB1	USB1-HS interface D1	1.8V	IO	
C13	PS_MIO29 USB1	USB1-HS interface DIR	1.8V		
A12	PS_MIO34 USB1	USB1-HS interface D2	1.8V	IO	
E13	PS_MIO38 USB1	USB1-HS interface D6	1.8V	IO	
A11	PS_MIO36 USB1	USB1-HS clock signal	1.8V	IO	
F12	PS_MIO35 USB1	USB1-HS interface D3	1.8V	IO	
A10	PS_MIO37 USB1	USB1-HS interface D5	1.8V	IO	
	USB_PHY_DP USB	USB1-HS interface DP	1.8V	IO	J1-P19
	USB_PHY_DM USB	USB1-HS interface DM	1.8V	IO	J1-P21
	USB_PHY_ID USB	USB1-HS Port ID	1.8V		J1-P25
	USB_PHY_VBUS	USB	USB1-HS interface VBUS	1.8V	J1-P27
	VBUS_SW_EN USB	USB1-HS interface EN	1.8V		J1-P30

Table 6-3 USB interface PIN definition

6.4. Ethernet Interface

MYC-C7Z010-20-V2 core board supports 2 Gigabit Ethernet ports. Users need to design PHY chip circuit and Ethernet port.

The Gigabit Ethernet Controller (GEM) implements 10/100/1000 Mbps Ethernet

The MAC, compliant with the IEEE 802.3-2008 standard, is capable of operating in half-duplex or full-duplex mode at all three speeds.

The controller provides MDIO interfaces for PHY management. The PHYs can be controlled from any of the MDIO interfaces.

6.4.1. Pin Definition

MCU pin number		Default functionality	Functional Description	Level input/output	core board pinout	
C11	PS_MIO53_501 PHY		LINK_MDIO_DATA	1.8 VO		
A17	PS_MIO20_501 PHY		LINK_RGMII0_TXD3 1.8V I			
D10	PS_MIO19_501 PHY		LINK_RGMII0_TXD2 1.8V I			
B18	PS_MIO18_501 PHY		LINK_RGMII0_TXD1	1.8 VI		
E14	PS_MIO17_501 PHY		LINK_RGMII0_TXD0 1.8V I			
A19	PS_MIO16_501 PHY		LINK_RGMII0_TX_CLK 1.8V I			
F14	PS_MIO21_501 PHY		LINK_RGMII0_TX_EN 1.8V I			
A15	PS_MIO26_501 PHY		LINK_RGMII0_RXD3 1.8V O			
B17	PS_MIO22_501 PHY		LINK_RGMII0_RX_CLK 1.8V O			
C7	PS_POR_B_500 PHY		PS_PWRON_RESETn 1.8V O			
F15	PS_MIO25_501 PHY		LINK_RGMII0_RXD2 1.8V O			
A16	PS_MIO24_501 PHY		LINK_RGMII0_RXD1 1.8V O			
D11	PS_MIO23_501 PHY		LINK_RGMII0_RXD0 1.8V O			
D13	PS_MIO27_501 PHY		LINK_RGMII0_RX_EN 1.8V O			
C10	PS_MIO52_501 PHY		LINK_MDIO_CLK	1.8 VO		
	RGMII0_TRP0 PHY		RGMII0_TRP0	1.8 VO		J1-P7
	RGMII0_TRN0 PHY		PHY differential signal lines	1.8 VO		J1-P9
	RGMII0_TRP1 PHY		PHY differential signal lines	1.8 VO		J1-P10
	RGMII0_TRN1 PHY		PHY differential signal lines	1.8 VO		J1-P12
	RGMII0_TRP2 PHY		PHY differential signal lines	1.8 VO		J1-P13
	RGMII0_TRN2 PHY		PHY differential signal lines	1.8 VO		J1-P15
	RGMII0_TRP3 PHY		PHY differential signal lines	1.8 VO		J1-P16
	RGMII0_TRN3 PHY		PHY differential signal lines	1.8 VO		J1-P18

Table 6-4 Ethernet interface PIN definition

6.5. QSPI Interface

The MYC-C7Z010-20-V2 core board supports up to 2 QSPI controllers and supports master/slave mode. SPI signals include SPI_CLK, SPI_MOSI and SPI_MISO, when designing, first confirm the relationship between the master and slave devices, and then confirm MOSI. The core board is configured with one QSPI interface by default. If you want to use more QSPI interfaces, port, please consult the chip manual for configuration.

6.5.1. Pin Definition

MCU pin label	default function	Functional Description	Level input/output core board pinout			
B8	QSPI0_D0 QSPI	QSPI MOSI Signal	3.3V IO			
D6	QSPI0_D1 QSPI	QSPI MISO Signal	3.3V IO			
A7	QSPI0_CS0 QSPI	QSPI chip select signal	3.3V IO			
A5	QSPI0_SCK QSPI	QSPI Clock Signal	3.3V IO			

Table 6-5 QSPI interface PIN definition

6.6. JTAG Interface

The standard JTAG interface is 4-wire: TMS, TCK, TDI, TDO, which are mode selection, clock, and data input respectively. PL-JTAG interface clock PL-JTAG uses the JTAG port to interface with the XADC and uses JTAG clock TCK. JTAG boot mode is considered a slave mode boot and is always an unsafe boot Mode. The JTAG chain can be configured in two ways: cascade and independent.

6.6.1. Pin Definition

MCU pin label	default function	Functional Description	Level input/output core board pinout			
F9	JTAG_TCK JTAG	JTAG clock signal	3.3V IO		J1-P94	
G6	JTAG_TDI JTAG	JTAG input signal	3.3V IO		J1-P96	
F6	JTAG_TDO JTAG	JTAG output signal	3.3V IO		J1-P98	
J6	JTAG_TMS JTAG	JTAG chip select signal 3.3V IO			J1-P100	

Table 6-6 JTAG interface PIN definition

6.7. HDMI interface

MYC-C7Z010-20-V2 is designed with HDMI interface, High Definition Multimedia Interface is a digital video/audio

Interface technology is a dedicated digital interface suitable for image transmission, which can transmit audio and image signals at the same time.

The data transmission speed is 2.25GB/s, and there is no need for digital/analog or analog/digital conversion before signal transmission.

6.7.1. Pin Definition

MCU foot	Label default function		Functional Description	Level input/output core board pinout			
T14	IO_B34_LP5 HDMI		12-bit input pixel data bus 3.3V	I		J2-P31	
T15	IO_B34_LN5 HDMI		12-bit input pixel data bus 3.3V	I		J2-P33	
T16	IO_B34_LP9 HDMI		12-bit input pixel data bus 3.3V	I		J2-P27	
U17	IO_B34_LN9 HDMI		12-bit input pixel data bus 3.3V	I		J2-P29	
U18	IO_B34_LP12 HDMI		12-bit input pixel data bus 3.3V	I		J2-P28	
U19	IO_B34_LN12 HDMI		12-bit input pixel data bus 3.3V	I		J2-P30	
V16	IO_B34_LP13 HDMI		12-bit input pixel data bus 3.3V	I		J2-P9	
W16	IO_B34_LN13 HDMI		12-bit input pixel data bus 3.3V	I		J2-P11	
V16	IO_B34_LP18 HDMI		12-bit input pixel data bus 3.3V	I		J2-P24	
W16	IO_B34_LN18 HDMI		12-bit input pixel data bus 3.3V	I		J2-P26	
R16	IO_B34_LP19 HDMI		12-bit input pixel data bus 3.3V	I		J2-P23	
R17	IO_B34_LN19 HDMI		12-bit input pixel data bus 3.3V	I		J2-P25	
T17	IO_B34_LP20 HDMI		12-bit input pixel data bus 3.3V	I		J2-P18	
R18	IO_B34_LN20 HDMI		12-bit input pixel data bus 3.3V	I		J2-P20	
V17	IO_B34_LP21 HDMI		12-bit input pixel data bus 3.3V	I		J2-P14	
V18	IO_B34_LN21 HDMI		12-bit input pixel data bus 3.3V	I		J2-P16	
W18	IO_B34_LP22 HDMI		12-bit input pixel data bus 3.3V	I		J2-P13	
W19	IO_B34_LN22 HDMI		12-bit input pixel data bus 3.3V	I		J2-P15	
N17	IO_B34_LP23 HDMI		12-bit input pixel data bus 3.3V	I		J2-P17	
P18	IO_B34_LN23 HDMI		12-bit input pixel data bus 3.3V	I		J2-P19	
V13	IO_B34_LN3 HDMI		Interrupt output	3.3V 0		J2-P52	
D8	PS_MIO7	HDMI	Reset signal	3.3 V	I	J1-P133	

Table 6-7 HDMI interface PIN definition

6.8. LCD Interface

MYC-C7Z010-20-V2 is designed with a 2x25_2.54mm pitch double-row pin interface.

It has multiple GPIO ports, LCD signals, FPGA signals, 3.3V, DC_IN_5V. The interface diagram is as follows.

6.8.1. Pin Definition

MCU pin label	default function		Functional Description	Level	input/output core	board pinout	
T14	IO_B34_LP5 HDMI		Display data signal 3.3V	IO		J2-P31	
T15	IO_B34_LN5 HDMI		Display data signal 3.3V	IO		J2-P33	
T16	IO_B34_LP9 HDMI		Display data signal 3.3V	IO		J2-P27	
U17	IO_B34_LN9 HDMI		Display data signal 3.3V	IO		J2-P29	
U18	IO_B34_LP12 HDMI		Display data signal 3.3V	IO		J2-P28	
U19	IO_B34_LN12 HDMI		Display data signal 3.3V	IO		J2-P30	
V16	IO_B34_LP13 HDMI		Display data enable 3.3V	IO		J2-P9	
W16	IO_B34_LN13 HDMI	display pixel	clock signal 3.3V	IO		J2-P11	
V16	IO_B34_LP18 HDMI		Display data signal 3.3V	IO		J2-P24	
W16	IO_B34_LN18 HDMI		Display data signal 3.3V	IO		J2-P26	
R16	IO_B34_LP19 HDMI		Display data signal 3.3V	IO		J2-P23	
R17	IO_B34_LN19 HDMI		Display data signal 3.3V	IO		J2-P25	
T17	IO_B34_LP20 HDMI		Display data signal 3.3V	IO		J2-P18	
R18	IO_B34_LN20 HDMI		Display data signal 3.3V	IO		J2-P20	
V17	IO_B34_LP21 HDMI		Display field signal 3.3V	IO		J2-P14	
V18	IO_B34_LN21 HDMI		Display line signal 3.3V	IO		J2-P16	
W18	IO_B34_LP22 HDMI		Display data signal 3.3V	IO		J2-P13	
W19	IO_B34_LN22 HDMI		Display data signal 3.3V	IO		J2-P15	
N17	IO_B34_LP23 HDMI		Display data signal 3.3V	IO		J2-P17	
P18	IO_B34_LN23 HDMI		Display data signal 3.3V	IO		J2-P19	
P15	IO_B34_LP24 HDMI		Capacitive touch reset 3.3V	VO		J2-P10	
P16	IO_B34_LN24 HDMI		Capacitive touch interrupt 3.3V	VO		J2-P12	
T19	IO_B34_25 HDMI	Undefined, high level valid	3.3V	IO		J2-P7	
R19	IO_B34_0	HDMI display backlight power	PWM control 3.3V	VO		J2-P8	

Table 6-8 LCD interface PIN definition

6.9. FMC Interface

MYC-C7Z010-20-V2 designs the FMC interface. FMC provides standard

Mezzanine board (daughter card) size, connectors and module interfaces. By separating the I/O interface from the FPGA in this way, there is no need to

It not only simplifies the I/O interface module design, but also maximizes the reuse rate of the carrier card. This model is ASP-134603-01,

and the specification is Array Socket SKT 160Pin 1.27mm Solder. It has compatibility: Standard

Optimized power supply, standardized signal definition, and increased compatibility with each other.

6.9.1. Pin Definition

Due to the large number of pins, we will not describe them in detail here. For detailed pin information, please refer to their specifications.

Book and schematics.

MCU pin label	default function		Functional Description	Level input/output core board pinout		
F9	JTAG_TCK FMC		FMC_JTAG clock signal 3.3V IO			
J6	JTAG_TMS FMC		FMC_JTAG mode select 3.3V IO			
G6	JTAG_TDI FMC		FMC_JTAG signal input 3.3V I			
F6	JTAG_TDO FMC		FMC_JTAG signal output 3.3V O			
	JTAG_TRST FMC		FMC_JTAG_TRST reset signal 3.3V IO			
T11	IO_B34_LP1 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J2-P61		
T10	IO_B34_LN1 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J2-P63		
T12	IO_B34_LP2 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J2-P62		
U12	IO_B34_LN2 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J2-P64		
U13	IO_B34_LP3 FMC		FMC_PRSENT signal pin 3.3V IO	J2-P57		
C20	IO_B35_LP1 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J1-P45		
B20	IO_B35_LN1 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J1-P47		
B19	IO_B35_LP2 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J1-P71		
A20	IO_B35_LN2 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J1-P73		
E17	IO_B35_LP2 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J1-P67		
D18	IO_B35_LN2 Differential clock signal	from FMC daughter card to motherboard	3.3V IO	J1-P69		

Table 6-9 FMC interface PIN definition

6.10. FPC interface

MYC-C7Z010-20-V2 is designed with an FPC connector. The connector model is FH12-50S-0.5SV, with a specification of The format is 50Pin, 0.5mm, 3.9mm Height. FPC has the characteristics of small size, light weight and thin thickness.

6.10.1. Pin Definition

MCU Pin Number	Default Function	Function	Description	Level	input/output	core board pinout	
T14	IO_B34_LP5	HDMI display data	signal 3.3V IO				
T15	IO_B34_LN5	HDMI display data	signal 3.3V IO				
T16	IO_B34_LP9	HDMI display data	signal 3.3V IO				
U17	IO_B34_LN9	HDMI display data	signal 3.3V IO				
U18	IO_B34_LP12	HDMI display data	signal 3.3V IO				
U19	IO_B34_LN12	HDMI display data	signal 3.3V IO				
V16	IO_B34_LP13	HDMI display data	enable 3.3V IO				
W16	IO_B34_LN13	HDMI display pixel	clock signal 3.3V IO				
V16	IO_B34_LP18	HDMI display data	signal 3.3V IO				
W16	IO_B34_LN18	HDMI display data	signal 3.3V IO				
R16	IO_B34_LP19	HDMI display data	signal 3.3V IO				
R17	IO_B34_LN19	HDMI display data	signal 3.3V IO				
T17	IO_B34_LP20	HDMI display data	signal 3.3V IO				
R18	IO_B34_LN20	HDMI display data	signal 3.3V IO				
V17	IO_B34_LP21	HDMI	Display field signal 3.3V IO				
V18	IO_B34_LN21	HDMI	Display line signal 3.3V IO				
W18	IO_B34_LP22	HDMI display data	signal 3.3V IO				
W19	IO_B34_LN22	HDMI display data	signal 3.3V IO				
N17	IO_B34_LP23	HDMI display data	signal 3.3V IO				
P18	IO_B34_LN23	HDMI display data	signal 3.3V IO				
P15	IO_B34_LP24	HDMI capacitive touch	reset 3.3VO				
P16	IO_B34_LN24	HDMI capacitive touch	interrupt 3.3VO				
T19	IO_B34_25	HDMI Undefined,	high level valid 3.3V IO				
R19	IO_B34_0	HDMI	Display backlight power PWM control system	3.3 VO			

Table 6-10 FPC interface PIN definition

6.11. RTC battery holder interface

MYC-C7Z010-20-V2 is designed with a button battery holder. The battery holder is 2Pin x 1.25mm spacing. The rated output voltage is 3.0V. It is controlled by I2C.

6.11.1. Pin Definition

MCU pin label	default function		Functional Description	Level	input/output	core board pinout	
B13	PS_501_MY_50	CAN	RTC_I2C_SCL	3.3 VI		J1-P126	
B9	PS_501_MY_51	CAN	RTC_I2C_SDA	3.3 VO		J1-P128	

Table 6-11 RTC clock PIN definition

6.12. CAN interface

MYC-C7Z010-20-V2 supports 2-way CAN controller. Complies with ISO 11898-1, CAN 2.0A, CAN2.0B standard. Supports standard frame (11-bit identifier) and extended frame (29-bit identifier). Supports 1mb/s bit rate.

6.12.1. Pin Definition

MCU Pin	Number	Default Function	Function	Description	Level	input/output	core board pinout
D5	PS_MIO8	CAN	CAN_TXD		3.3 VI		J1-P113
B5	PS_MIO9	CAN	CAN_RXD		3.3 VO		J1-P115

Table 6-12 CAN interface PIN definition

6.13. GPIO Interface

The General Purpose I/O (GPIO) peripheral provides software observation and control of up to 54 device pins through the MIO module.

It also provides access to 64 inputs from the programmable logic (PL) and 128 outputs to the PL through the EMIO interface.

The GPIOs are organized into four groups of registers that group related interface signals. Each GPIO acts as

Input, output or interrupt sensing is independently and dynamically programmable. Software can read the bank using a single load instruction.

All GPIO values, or write data to one or more GPIOs (within the range of the GPIO) using a single store instruction. The GPIO control and status registers are memory mapped at base address 0xE000 A000.

6.13.1. Pin Definition

The GPIO module is divided into four groups: Bank0: 32-bit bank controls MIO pins [31:0]

Bank1: 22-bit bank controls MIO pins [53:32]; Note: Bank1 is limited to 22 bits because MIO has a total of 54 pins.

Bank2: 32-bit bank controls EMIO signal [31:0]; Bank3: 32-bit bank controls EMIO signal [63:32] Software controls the GPIO through a series of

memory-mapped registers. The control of each bank is the same, although there are some minor differences between the MIO and EMIO banks due to their different functions.

6.14. Description of the core board internal resource pins

The MYC-C7Z010-20-V2 core board uses some IO resources, and these pins have been set to fixed functions.

Yes, you cannot change the resource assignments of these pins.

In addition, considering that users may want to define core board resources more freely when developing products based on the Mir core board,

When defining pin resources, please exclude the following pins and do not overlap the following default functions when defining other pins.

complex.

MCU Pinout	Default functionality	Remark
E9	eMMC_D0	EMMC
E8	eMMC_D1	
C5	eMMC_D2	
C8	eMMC_D3	
C6	eMMC_CMD	
D9	eMMC_CLK	
U13	IO_B34_LP3	LED indicators
R11	FPGA_CONFIG_DONE LED indicator	D7
E6	WDT_FEED	Watchdog
D8	PS_500_RESET_OUTn Reset	
A19	RGMII_TX_CLK	Network Chip
E14	RGMII_TXD0	Network Chip
B18	RGMII_TXD1	Network Chip
D10	RGMII_TXD2	Network Chip
A17	RGMII_TXD3	Network Chip
F14	RGMII_TX_EN	Network Chip
B17	RGMII_RX_CLK	Network Chip
D11	RGMII_RXD0	Network Chip
A16	RGMII_RXD1	Network Chip
F15	RGMII_RXD2	Network Chip
A15	RGMII_RXD3	Network Chip

D13	RGMII_RX_EN	Network Chip
C10	MDIO_CLK	Network Chip
C11	MDIO_DATA	Network Chip
MIO09		Not used
MY50		Not used
MIO51		Not used

Table 6-14 Core board internal resource pin definition

7. Packaging information

7.1. Mechanical dimensions

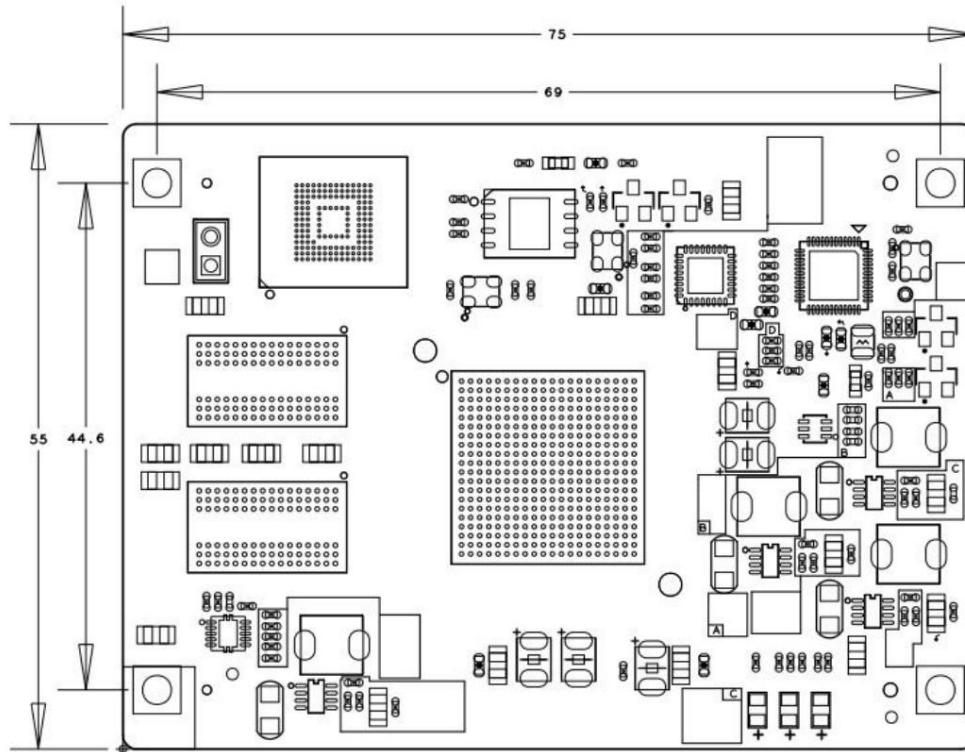


Figure 7-1 MYC-C7Z010-20-V2 core board top view (unit: mm)

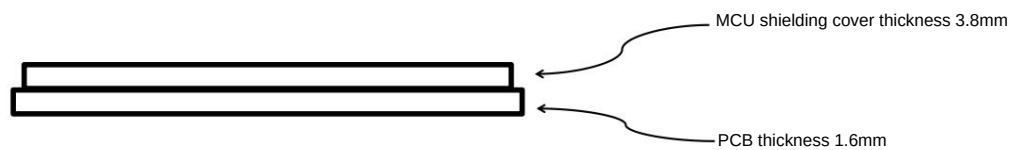


Figure 7-2 Side view of the MYC-C7Z010-20-V2 core board (this version has no shielding cover)

7.2. Base PCB packaging

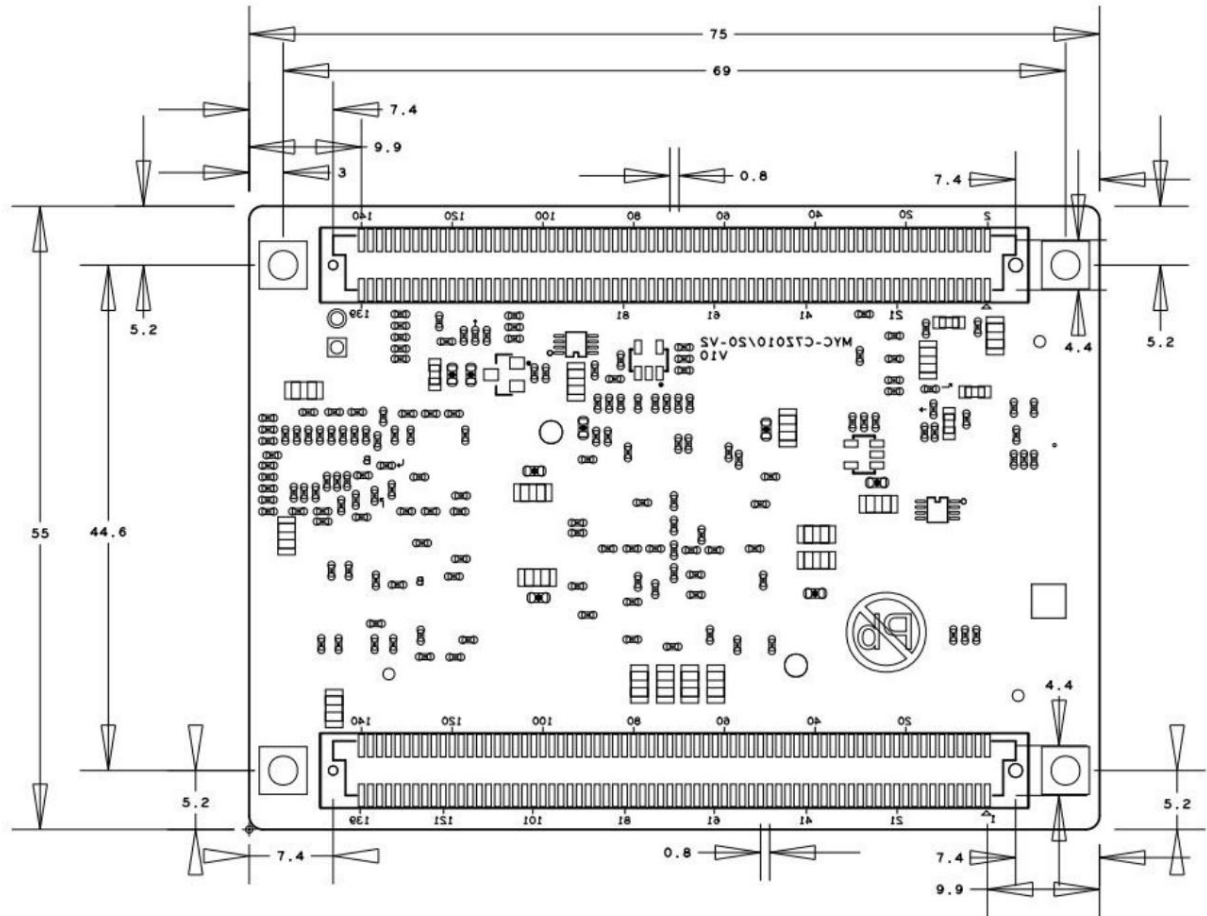


Figure 7-3 MYC-C7Z010-20-V2-Core Board PCB Package (Unit: mm)

Connector pins 140Pin pitch 0.8mm. Mir Electronics provides designed PCB packaging, please visit <http://down.myr-tech.com/> to obtain the package file.

7.3. Baseboard PCB Requirements

- a) The recommended PCB thickness is at least 1.6mm. Pay attention to the balance of copper coating. If the PCB is deformed during the furnace, it is recommended to use a carrier to fix the furnace. b) To ensure the quality of mounting and tinning, please ensure that the distance between the module and other components on the PCB is at least 3mm. c) Please design the package of the core board module according to Section 7.2, or use the PCB package provided by Mir Electronics.

8. Mounting and storage requirements

8.1. Steel mesh design

a) Thickness: The recommended steel mesh thickness for the module pad is 0.18-0.2mm. b) Opening: If the recommended pad is used, the width of the steel mesh opening should be 10% smaller than the pad, and the length should be 15% longer than the pad. If the recommended pad is not used, it is recommended to open the corresponding pad at a ratio of 1:1.2, and pay attention to avoid continuous tin.

Note: If the stencil thickness cannot be designed as recommended, resulting in insufficient tin, consider adding preformed solder paste.

8.2. Storage requirements

The module is shipped in a vacuum-sealed form and must be stored under the following conditions:

a) The vacuum-sealed bag can be stored for 12 months when the ambient temperature is below 40°C and the air humidity is less than 90%. b) After the vacuum-sealed bag is opened, reflow soldering can be performed directly within 72 hours when the ambient temperature is below 30°C and the air humidity is less than 10%. Note: If the above conditions are not met, baking should be performed before mounting.

8.3. Baking method

Since the module packaging material cannot withstand high temperatures, if necessary, please choose one of the following two methods to bake to avoid affecting the module welding quality. a) Original packaging baking: baking temperature is 40-60°C, time is 5-7 days. b) Transfer to high temperature resistant tray for baking: baking temperature is 100-120°C, baking time is more than 48 hours. **8.4. Welding process**

a) If the substrate to be mounted is a double-sided device layout, it is recommended to place the core board mounting process in the last stage. b) It is recommended to set the time of the preheating area (160-200°C) to 60-120 seconds. c) The recommended reflow temperature is 235-245°C, and the maximum temperature should not exceed 250°C. The reflow time is recommended to be controlled within 40-60 seconds. d) The recommended temperature rise rate is 1-3°C/second, and the temperature drop rate is 2-4°C/second.

Appendix 1 Contact Us

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When you get help via email, please use the following format to write the email title so that the corresponding development team can quickly follow up and handle your problem.

question:

[Company Name/Individual--Development Board Model] Problem Overview

Appendix 2 After-sales service and technical support

All Mir Technology products purchased directly from Mir Technology or from authorized Mir Technology agents are

You can enjoy the following benefits:

1. 6-month free warranty service cycle 2. Lifetime free technical support service 3. Lifetime repair service 4. Free software upgrade service for the purchased product 5. Free software source code for the purchased product and some software source code developed by Mir Technology
6. You can purchase samples of major chips directly from Mir Technology, which is simple, convenient and fast; you don't have to wait for a long time when purchasing from agents.
7. From the date of purchase, you will become a permanent customer of Mir Technology and enjoy preferential policies for purchasing any software or hardware products of Mir Technology again.
8. OEM/ODM service

If any of the following circumstances occurs, the free warranty service will not be provided:

1. Expiration of the free warranty service period 2. No product serial number or valid product purchase receipt
3. Liquid, moisture, mold or corrosion
4. Failures and damages caused by impact, squeezing, falling, scratching, etc. that are not quality issues of the product itself
5. Failures and damages caused by unauthorized hardware modification, incorrect power-on, or incorrect operation
6. Failure and damage caused by force majeure

Product Return

If the product fails, is damaged or has other abnormalities during use, please call Mir Technology Customer Service before sending it back for repair.

Department, communicate with engineers to confirm the problem to avoid unnecessary freight losses and cycle delays caused by misjudgment of the fault.

Maintenance cycle

After receiving the returned product, we will arrange engineers to inspect it immediately. We will repair or replace it and send it back in the shortest possible time. The general fault repair cycle is 3 working days (from the date our company receives the item, excluding the transportation process time). For products that cannot be repaired in a short period of time due to special faults, we will communicate with the user separately and confirm the repair cycle.

Repair costs

For products within the free warranty period, no repair fee will be charged for failures caused by product quality problems;

If the product fails or is damaged within the warranty period, we will communicate with the customer and confirm the repair fee after testing and confirming the problem. We only charge the component material fee, and do not charge the repair service fee. For products that exceed the warranty period, the component material fee and repair service fee will be determined based on the actual degree of damage.

Shipping costs

When the product is under normal warranty, the user shall bear the shipping fee for returning the product, and our company shall bear the cost of returning the product to the user after repair.

The shipping costs of the products are borne by the user.